CATS^{+}

Competishun Advance

Training Sessions

(Lecture - XIV)

Ans

Let f(x) be a thrice differentiable function defined on $(-\infty, \infty)$ such that

$f(x) = f(2-x)$ and $f'\left(\frac{1}{2}\right) = f'\left(\frac{1}{4}\right) = 0$ Then which of the following is are/true

(A) Minimum number of values where $f''(x)$ vanishes on $[0, 2]$ is 4

(B) $\int_{-1}^{1} f'(1+x)\, x^2 e^{x^2}\, dx = 0$

(C) $\int_{0}^{1} f(1-t)\, e^{-\cos\pi t}\, dt - \int_{1}^{2} f(2-t)\, e^{\cos\pi t}\, dt = \int_{0}^{2} f'(t)\, e^{\cos\pi t}\, dt$

(D) There exist a 'x' such that $x \in \left(\frac{1}{4}, 1\right)$ where $f'''(x)$ vanishes.

A, B, C, D

(C)

$$\text{L.H.S.} = \int_0^1 f(1-t)\cdot e^{-\cos(\pi t)}\cdot dt - \int_1^2 f(2-t)\cdot e^{\cos(\pi t)}\cdot dt$$

$$t-1 = z$$
$$dt = dz$$

$$= \int_0^1 f(1-z)\cdot e^{-\cos(\pi z)} dz$$

$$= \underline{0}.$$

$$\underline{\text{R.H.S}} = \int_0^2 f'(t)\cdot e^{\cos(\pi t)}\cdot dt \quad (t \to 2-t)$$

$$= \int_0^1 \left(f'(t)\cdot e^{\cos(\pi t)} + f'(2-t)\cdot e^{\cos(\pi t)}\right) dt$$

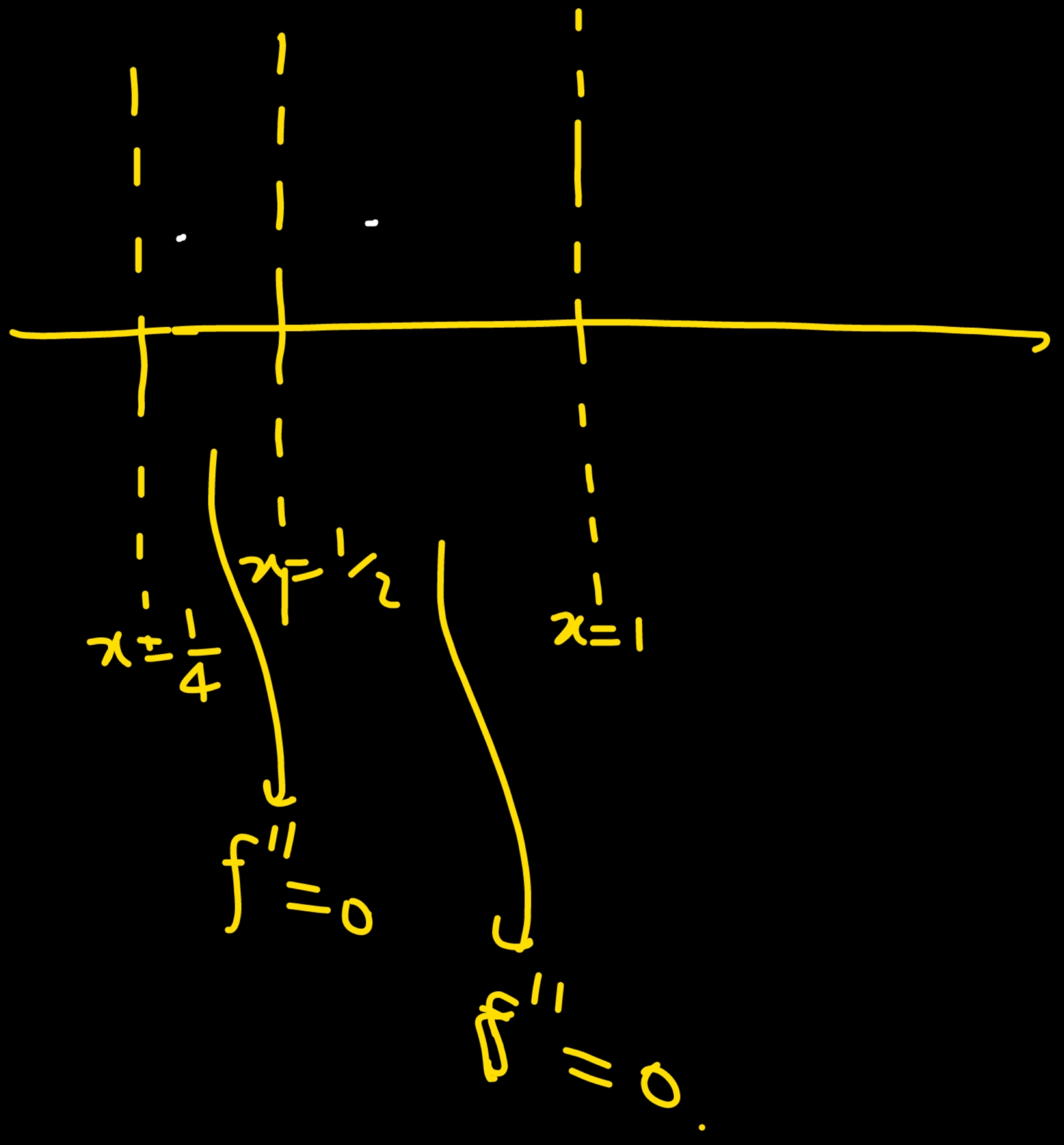
$$= \int_0^1 \left(\cancel{f'(t)+f'(2-t)}\right)^{0}\cdot e^{\cos(\pi t)}\cdot dt$$

$$= \underline{0}$$

Property

$$\int_0^{2a} f(x)\cdot dx = \int_0^a \left(f(x)+f(2a-x)\right) dx$$

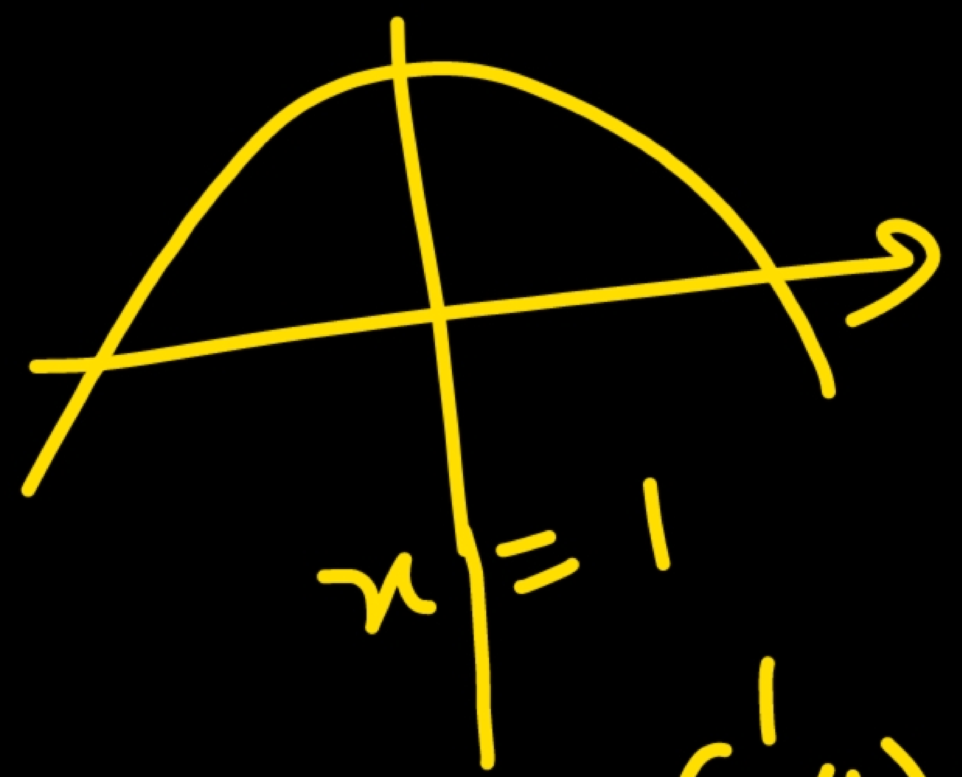
$$f(x) = f(2-x)$$
$$\xrightarrow{\frac{d}{dx}} f'(x) = -f'(2-x)$$
$$f'(x) + f'(2-x) = 0$$



f', f'' & $f''' \longrightarrow$ exists.

$f(x) = f(2-x)$

$\hookrightarrow x = 1$ symt



$f'(1) = 0$

$f'\left(\frac{1}{2}\right) = 0$

$f'\left(\frac{1}{4}\right) = 0$

(B) $$\int_{-1}^{1} f'(1+x)\cdot x^2\cdot e^{x^2}\cdot dx = \int_{0}^{1}\left(f'(1+x)\cdot x^2\cdot e^{x^2} + f'(1-x)\cdot x^2\cdot e^{x^2}\right)dx$$

$$= \int_{0}^{1}\left(\underbrace{f'(1+x) + f'(1-x)}_{=0}\right)\cdot x^2\cdot e^{x^2}\cdot dx$$

$$= 0.$$

$$\boxed{\int_{-a}^{a} f(x)\,dx = \int_{0}^{a}\left(f(x)+f(-x)\right)dx}$$

Given

$$f(x) = f(2-x)$$

$x \to 1+x$

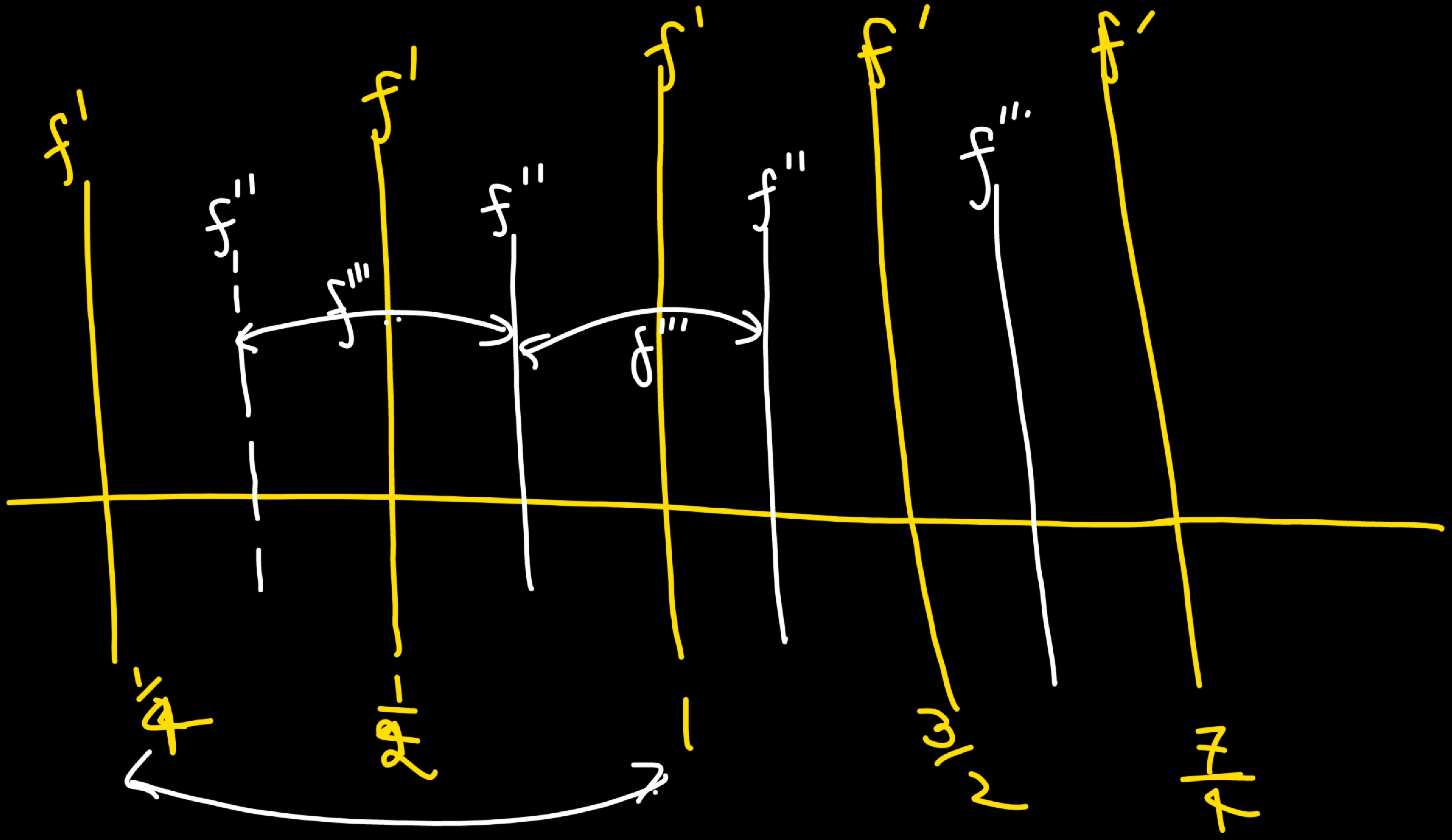
$$f(1+x) = f(1-x)$$

$\frac{d}{dx}$

$$f'(1+x) = -f'(1-x)$$

$$f'(1+x) + f'(1-x) = 0.$$

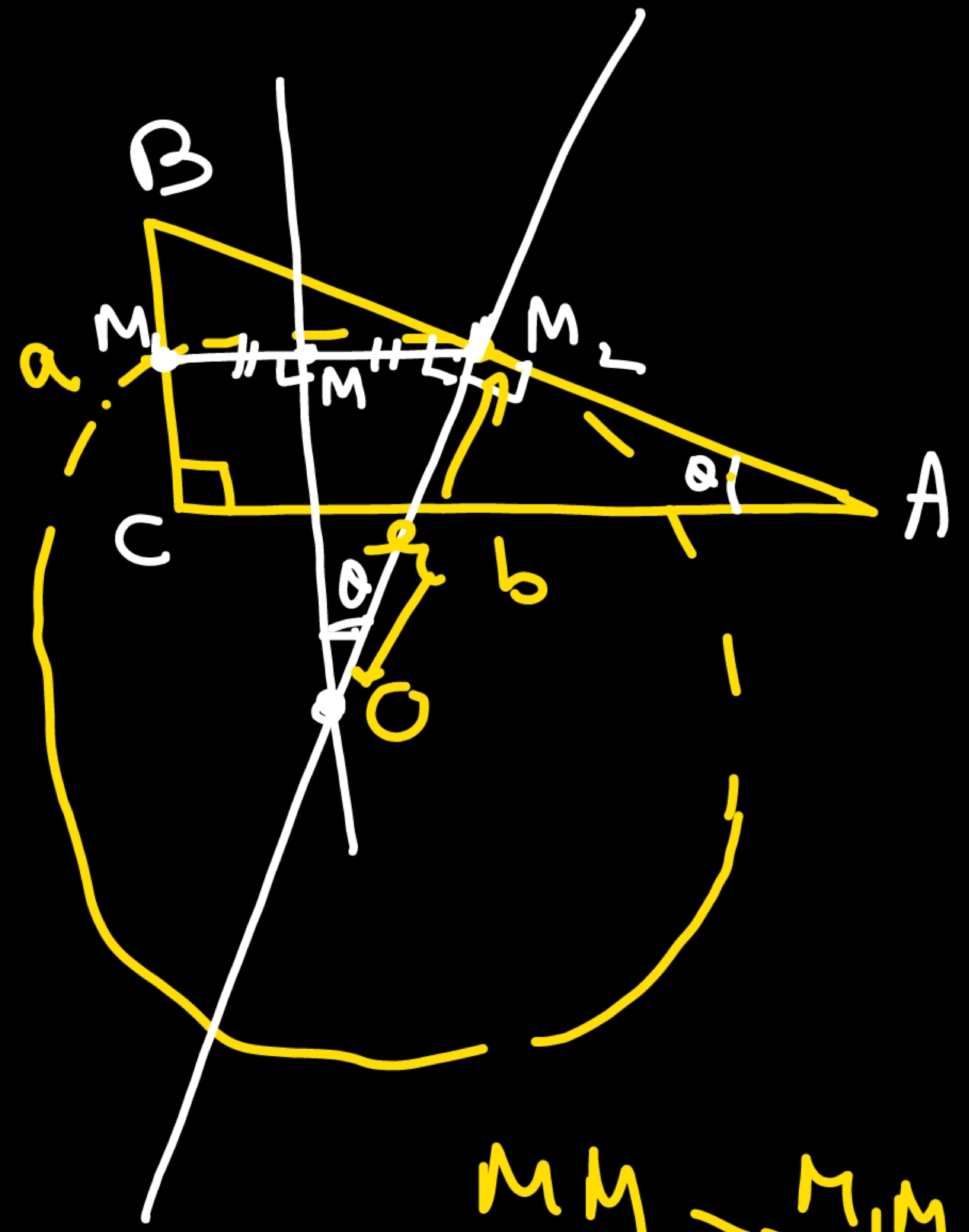
D



Que

A circle touches the hypotenuse of a right triangle at its middle point and passes through the mid point of the shorter side. If a and b (a< b) be the legs of the triangle then radius of the circle will be

(A) $\frac{a}{2b}\sqrt{a^2+b^2}$ (B) $\frac{b}{2a}\sqrt{a^2+b^2}$ (C) $\frac{a}{4b}\sqrt{a^2+b^2}$ (D) $\frac{b}{4a}\sqrt{a^2+b^2}$



In ΔMM_2O

$\sin(\theta) = \frac{MM_2}{OM_2}$

$\frac{a}{\sqrt{a^2+b^2}} = \frac{b/4}{r}$

$\boxed{r = \frac{b}{4a}\sqrt{a^2+b^2}}$

$M_1M_2 \parallel AC$

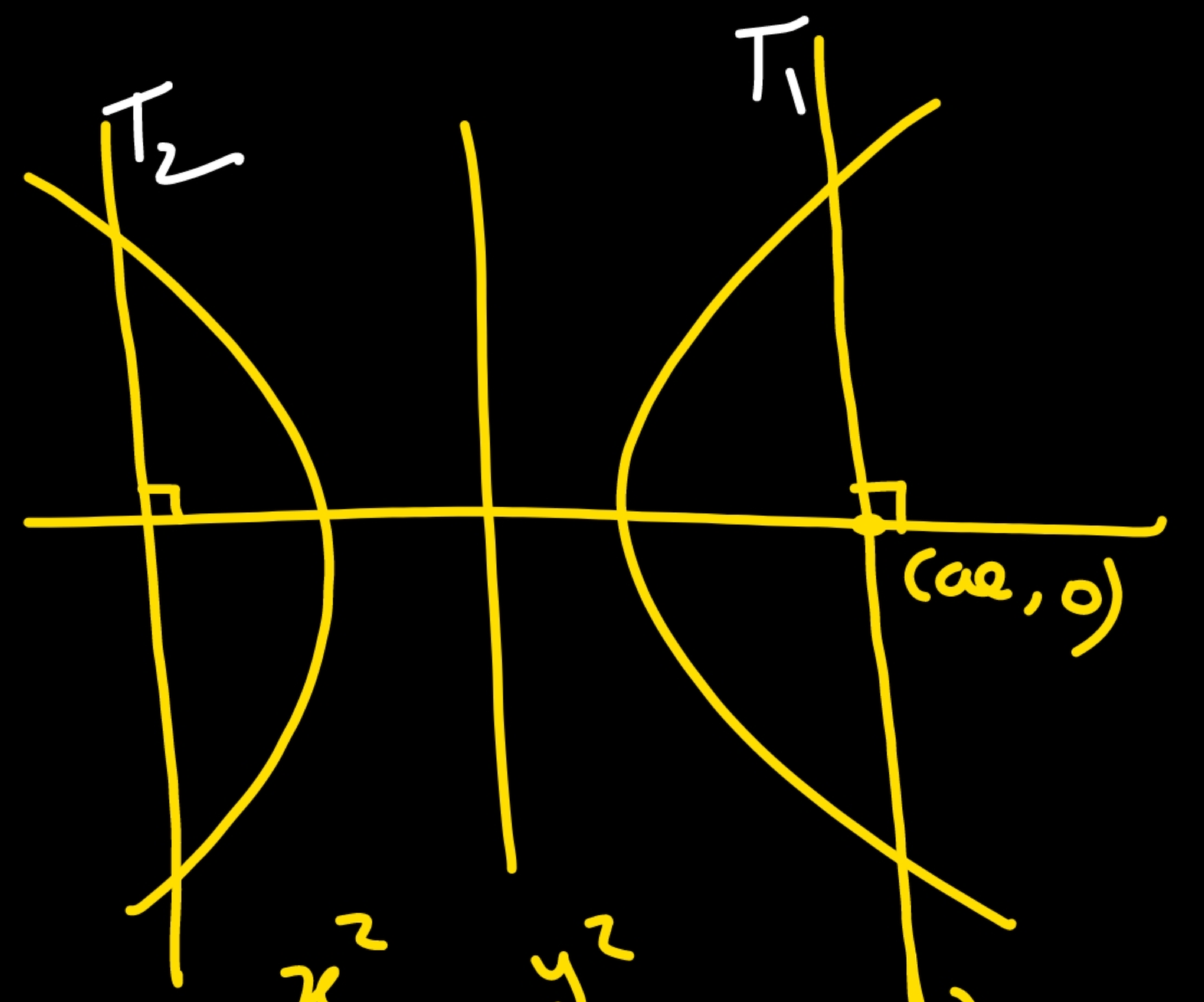
$M_1M_2 = \frac{AC}{2} = b/2.$

$MM_2 = \frac{M_1M_2}{2} = \frac{b}{4}.$

Que

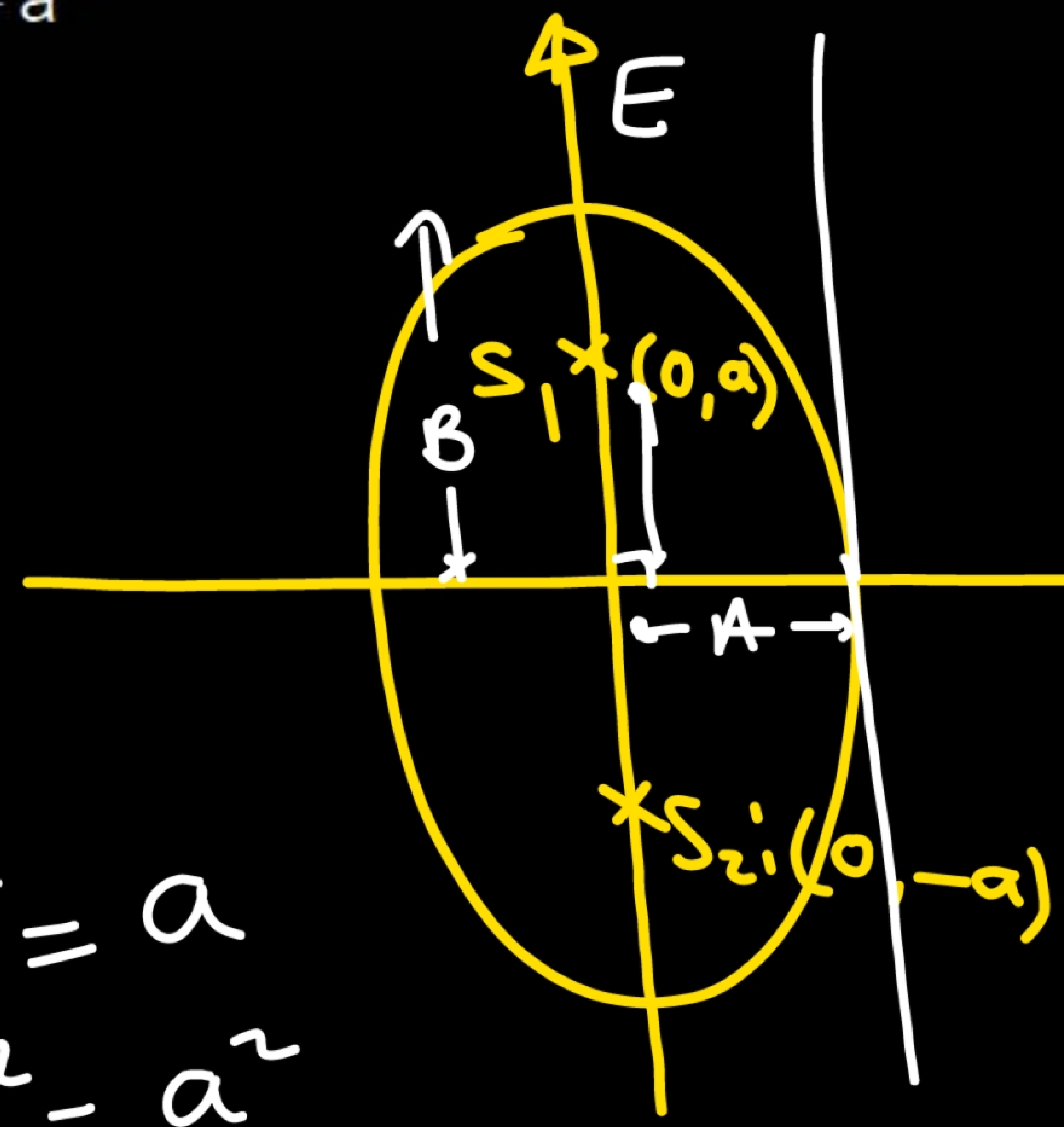
The equation of ellipse having latus rectum of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ as tangents and $(0, \pm a)$ as its foci is

(A) $\frac{x^2}{a^2} + \frac{y^2}{a^2+b^2} = 1$ (B) $\frac{x^2}{a^2+b^2} + \frac{y^2}{2a^2+b^2} = 1$ (C) $\frac{x^2}{b^2} + \frac{y^2}{2b^2+a^2} = 1$ (D) none of these



$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$e = \sqrt{1 + \frac{b^2}{a^2}}$



$B \cdot E = a$

$B^2E^2 = a^2$

$B^2 - A^2 = a^2$

$B^2 = A^2 + a^2$

$= 2a^2 + b^2$

$\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$

$E^2 = 1 - \frac{A^2}{B^2}$

$B^2E^2 = B^2 - A^2$

$A = ae$

$A^2 = a^2e^2$

$= a^2 + b^2$

$$\boxed{\frac{x^2}{a^2+b^2} + \frac{y^2}{2a^2+b^2} = 1}$$

Column Matching

Column - I

A. If $f(x) = \cos^{-1}\sqrt{\dfrac{\cos 3x}{\cos^3 x}}$ and $f'(x)$ can be expressed in the form of $\sqrt{\dfrac{p}{\cos qx + \cos rx}}$ (where p, q, r are natural numbers) given that $q > r$. Then find the value of $p + q - 2r$.

6

$p = 6$
$r = 2$
$q = 4$

B. Find minimum value of
$f(x) = x^8 + x^6 - x^4 - 2x^3 - x^2 - 2x + 9$.

C. Let $f : R \to R$ is continuous function satisfying the equation $f(x) + f\left(\dfrac{2x}{3}\right) = x$ and $f(0) = 0$. If $f(1) = \lambda$, then find the value of 10λ.

Column-II

P. 4

Q. 5

R. 6

S. 8

T. None of P, Q, R & S.

A → R

B → Q

C → R.

Ⓐ $$f(x) = \cos^{-1}\left(\sqrt{\frac{\cos(3x)}{\cos^3(x)}}\right) = \cos^{-1}\left(\sqrt{\frac{4c^3 - 3c}{c^3}}\right) = \cos^{-1}\left(\sqrt{4 - 3\sec^2(x)}\right)$$

$$f'(x) = -\frac{1}{\sqrt{1 - (4 - 3\sec^2(x))}} \cdot \frac{1}{\cancel{2}\sqrt{4 - 3\sec^2(x)}} \cdot \left(-3 \cdot \cancel{2} \cdot \sec^2(x) \cdot \tan(x)\right)$$

$$= \frac{3\sec^2(x) \cdot \cancel{\tan(x)}}{\sqrt{3\cancel{(\sec^2(x) - 1)}}} \cdot \frac{1}{\sqrt{4 - 3\sec^2(x)}}$$

$$= \sqrt{\frac{3}{\cos^4(x)\left(4 - 3\sec^2(x)\right)}} = \sqrt{\frac{3}{\cos(x)\left(4\cos^3(x) - 3\cos(x)\right)}}$$

$$= \sqrt{\frac{6}{2\cos(x)\cos(3x)}}$$

$$= \sqrt{\frac{6}{\cos(2x) + \cos(4x)}}$$

(B)

$$f(x) = x^8 + x^6 - x^4 - 2x^3 - x^2 - 2x + 9$$

$$f'(x) = 8x^7 + 6x^5 - 4x^3 - 6x^2 - 2x - 2$$

$$= (x-1)\cdot(8x^6 + 8x^5 + 14x^4 + 14x^3 + 10x^2 + 4x + 2)$$

$$= 2(x-1)(4x^6 + 4x^5 + 7x^4 + 7x^3 + 5x^2 + 2x + 1)$$

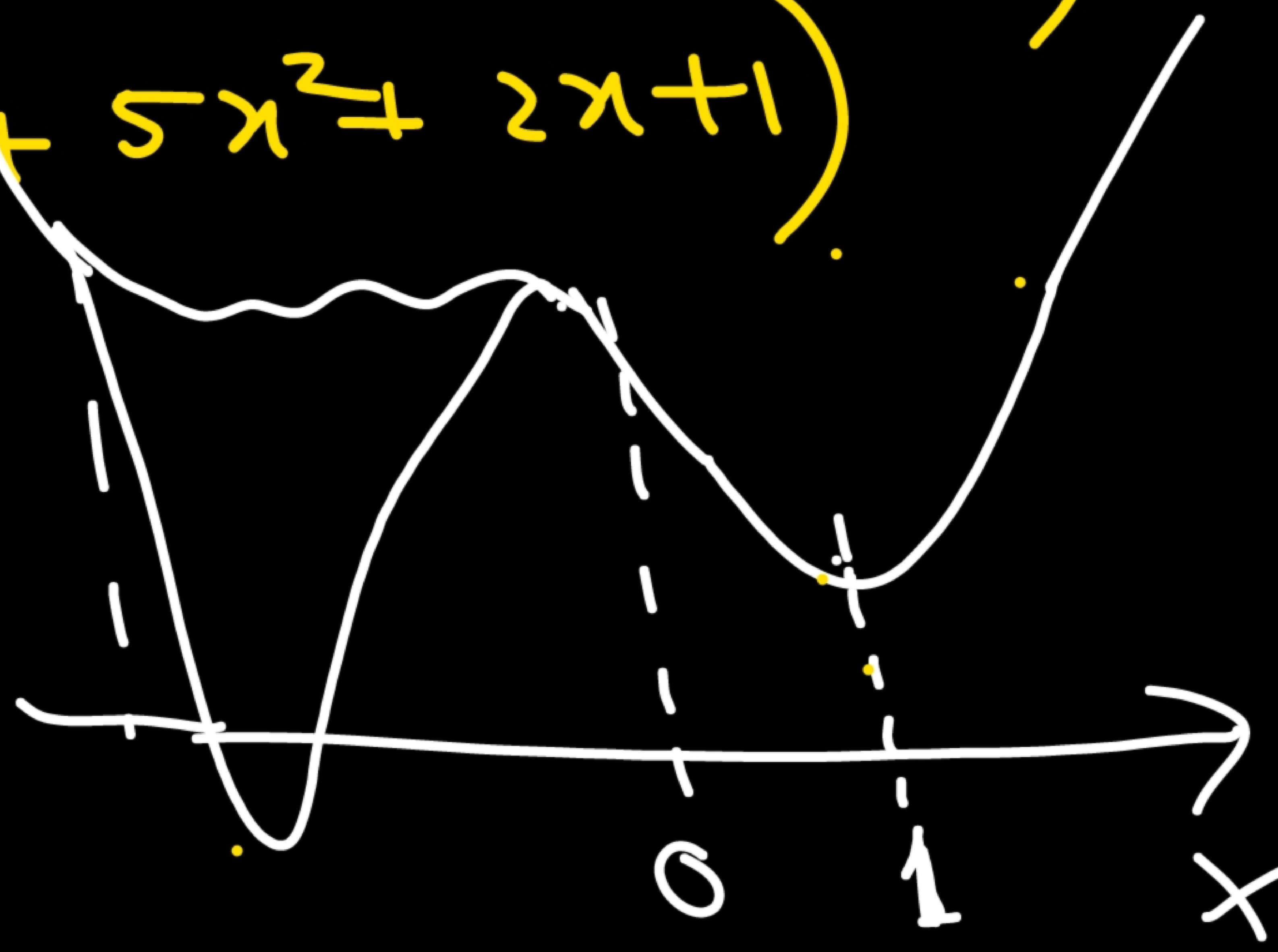
$$f_{min.} = f(1)$$

$$= 11 - 6$$

$$= \boxed{5}$$

sign of f'

+/-		−		+
	0		1	



ⓒ $f : R \to R$

$$f(x) + f\left(\tfrac{2}{3}x\right) = x$$

$f(0) = 0$

$f(1) = \lambda$

$10\lambda = ?$

$$\Rightarrow f(x) = x - f\left(\tfrac{2}{3}x\right)$$

$$= x - \left(\tfrac{2}{3}x - f\left(\tfrac{2}{3}\cdot\tfrac{2}{3}x\right)\right)$$

$$= x - \tfrac{2}{3}x + f\left(\tfrac{4}{9}x\right)$$

$$= x - \tfrac{2}{3}x + \tfrac{4}{9}x - f\left(\left(\tfrac{2}{3}\right)^3 x\right)$$

$\vdots$

$$f(x) = x\Big(\underbrace{1 - \tfrac{2}{3} + \left(\tfrac{2}{3}\right)^2 - \left(\tfrac{2}{3}\right)^3 + \cdots}_{n\text{ terms}}\Big) + (-1)^n \cdot f\Big(\underbrace{\left(\tfrac{2}{3}\right)^n x}_{\to 0}\Big)$$

$= f(0)$

$\downarrow n \to \infty$

$$f(x) = x\left(\frac{1}{1-\left(-\frac{2}{3}\right)}\right) + 0 = \frac{3}{5}x$$

$$\boxed{f(x) = \frac{3}{5}x} \xrightarrow{x=1} f(1) = \frac{3}{5} = \lambda$$

$$10\lambda = \boxed{6}$$

Que

Define a 'good word' as a sequence of letters that consists only of letters A, B and C and in which A is never immediately followed by B, C is never immediately followed by B. Let P_n be the probability of obtaining 'good word' from n-letter word then which of the following is/are true. (where $n \geq 3$)

(A) $P_n = \frac{2}{3}P_{n-1} + \frac{1}{3}$ (B) $P_4 = \frac{73}{81}$ (C) $P_5 = \frac{227}{243}$ (D) $P_5 = \frac{7}{27}$

nth letter

X ABACA - - -

AAAC - - - -

CCCCA - - - -

BAAC - - - -

P_{n-1}

A/B/C | A

B - - - - B B | B

A/B/C | C

$$P_n = P_{n-1}\cdot\frac{1}{3} + \left(\frac{1}{3}\right)^n + P_{n-1}\cdot\frac{1}{3}$$

$$\boxed{P_n = \frac{2}{3}P_{n-1} + \left(\frac{1}{3}\right)^n}$$

$$P_n = \frac{2}{3} \cdot P_{n-1} + \left(\frac{1}{3}\right)^n$$

$P_1 = 1$

$\downarrow n = 2.$

A

B

C

$$P_2 = \frac{2}{3} \cdot 1 + \frac{1}{9} = \frac{7}{9}$$

$n=3$

$$P_3 = \frac{2}{3} \cdot \frac{7}{9} + \left(\frac{1}{3}\right)^3 = \frac{15}{27} = \frac{5}{9}$$

$n=4$

$$P_4 = \frac{2}{3} \cdot \frac{5}{9} + \frac{1}{81} = \frac{31}{81}$$

$n=5$

$$P_5 = \frac{2}{3} \cdot \frac{31}{81} + \frac{1}{243} = \frac{\cancel{63}\,7}{\cancel{243}\,27}$$

$$= \frac{7}{27}$$

Qn

A matrix has nine elements, in which seven elements are 0 and two elements are 1, then

~~(A)~~ Probability of getting a square diagonal matrix is $\frac{1}{18}$

~~(B)~~ Probability of getting a singular square matrix is $\frac{1}{2}$

(C) Probability of getting a square matrix whose trace is zero is $\frac{5}{36}$

(D) Probability of getting a square matrix is $\frac{1}{3}$

C D

Sol

$7 \longrightarrow 0\text{'s}$

$2 \longrightarrow 1\text{'s}$

Order $\longrightarrow$ $\underline{3\times 3}$, 9×1, 1×9.

$\frac{1}{3}$

$\frac{1}{3} \times \frac{{}^3C_2}{{}^9C_2}$

$= \frac{1}{\cancel{3}} \times \frac{\cancel{2}}{9\times 8} = \frac{1}{72}$.

$\begin{bmatrix} & 0 & 0 \\ 0 & & 0 \\ 0 & 0 & \end{bmatrix}$

C

$\frac{1}{3} \times \frac{{}^6C_2}{{}^9C_2} = \frac{1}{\cancel{3}} \frac{\cancel{6}\times 5}{9\times \cancel{8}\,4} = \boxed{\frac{5}{36}}$